

ADAM FARIDI

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EDUCATION

Cornell University

B.S. Computer Science & Operations Research and Engineering (Double Major)

GPA: 3.41/4.0

May 2025

Relevant Coursework: Analysis of Algorithms, Deep Learning, Financial Engineering, Machine Learning, Optimization (Convex and Non-Convex), Real Analysis (Honors), Object-Oriented Programming, Data Structures, Engineering Probability and Statistics.

PROFESSIONAL EXPERIENCE

Mastercard

Data Engineer I

St. Louis, MO

August 2025 - Present

- Architected, implemented, and deployed production-scale data pipelines that power **100+** BI dashboards via HTTP requests to REST APIs, Python/SQL programming, and ETL/ELT tools like Apache Airflow and Alteryx.
- Implemented large-scale data processing and statistical analysis workflows in Python and SQL over **Hadoop**, analyzing **billions** of transaction records for anomaly and trend detection.
- Collaborated on developing an **AI agent** for Mastercard's internal IT tool to automate operations processes.

Data Engineer Intern

June 2024 - August 2024

- Developed an LDA-based topic modeling pipeline in Python to extract latent themes from 67,000+ IT tickets, enabling large-scale analysis of operational trends.
- Integrated an **LLM** into a production-level data pipeline to perform complex natural language data cleaning tasks.

Paramount

Data Science & Research Intern for CBS News

New York, NY

June 2023 - August 2023

- Implemented a logistic **regression model**, providing finance with viewership forecasts for **5** CBS News programs.
- Conducted statistical analysis of daily and weekly trends in viewership and market share relative to NBC and ABC.
- Built an automation pipeline with Python and SQL modules to automate daily analytics reports and press releases.

Labor Dynamics Institute at Cornell

Research Assistant

Ithaca, NY

January 2023 - May 2025

- Orchestrated high-performance computing workflows for empirical research using economic and financial datasets, including **Bloomberg** terminal data, leveraging Python, Unix, Git, Docker, and distributed job scheduling.
- Executed regressions, ML models, and multi-week simulations for 25 publications in the American Economic Review.

PROJECTS & RESEARCH

Independent Research under Cornell ORIE Faculty | CI/CD, PyTorch, SciPy, NumPy, Optimization Algorithms, Proofs

- Contributed to the open-source *SimOpt* package by writing testing and analysis code for its optimization solvers, identifying a flaw in a solver used in prior literature through empirical and theoretical validation.
- Conducted research on convexity properties of nonlinear simulation optimization problems.
- Analyzed convergence behavior and correctness of various optimization algorithms.

Research Paper for Revenue Optimization | Python, Mathematical Programming, Proofs, Optimization, GUROBI, SciPy

- Derived theoretical bounds on the performance gap between revenue optimization models using mathematical programming and proof-based analysis, in collaboration with a WashU professor (paper forthcoming Summer 2026).

Fine-tuning Text-To-Image Diffusion | Python, GCP, MLFlow, TensorFlow, HuggingFace, BeautifulSoup

- Reimplemented the DreamBooth diffusion research paper to personalize text-to-image models, achieving similar results.
- Implemented Prior Preservation Loss (PPL), which prevented overfitting to subject-specific data in fine-tuned models.
- Established a GCP-based training environment to support our model training and data augmentation workflows.
- Presented results as a technical poster in a team of 5 to 150+ students in a graduate-level deep learning course.

LEADERSHIP & ACTIVITIES

Teaching Assistant for INFO 3950 – Data Analytics for Information Science

Teaching Assistant

Ithaca, NY

January 2025 – May 2025

- Mentored **30** students in data science and ML programming by holding weekly office hours, 1-on-1 tutoring, and grading.

Cornell Seismic Design (CSD)

Construction Team Lead

Ithaca, NY

August 2021 – May 2025

- Led the construction of a 5-foot wooden tower, placing **2nd**/38 in April 2025 at an annual global seismic competition.

SKILLS & INTERESTS

Skills: Python, C/C++, SQL, Java, R, OCaml; NumPy, SciPy, Pandas, PyTorch, TensorFlow, Gurobi; Hadoop, PostgreSQL, MySQL, Apache Airflow, Alteryx; Linux, Docker, Git, AWS, CI/CD, MLflow; Time-Series Analysis, Optimization & Statistical Modeling.

Hobbies: Architecture, Urban Planning, Soccer, Geography, Language Learning (French & Japanese), Gym/Fitness, Stunt Teams